

THEORY AND METHODS NEW VERSIONS OF THE EXTREMAL TYPES THEOREM

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Summary: This paper deals with the continuation property of the weak convergence $\left(\xrightarrow{w}\right)$ of extreme order statistics. Under general conditions, the Extremal Types Theorem is proved, when the convergence of the distribution functions (d.f.'s) of the extremes is restricted on a countable set of real values. Furthermore, under these conditions, the convergence is investigated when the sample size is a positive integer valued random variable (r.v.) v_n .

1. Introduction and the statement of the problem

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed r.v.'s with a common continuous d.f. F and write $X_{r:n}$ for r th largest of the first n , i.e. $X_{r:n}$ is the r th order statistic (o.s.). All results obtained for $X_{n-r+1:n}$ of course lead to analogous results for $X_{r:n}$. We shall therefore only discuss $X_{n-r+1:n}$ in this work, where r is a positive fixed integer. The possible forms for the limiting d.f. of $X_{n-r+1:n}$, under linear normalizations, are completely specified by the Extremal Types Theorem (see Arnold *et al.*, 1992 and Leadbetter *et al.*, 1983). More specifically, this basic classical result

states that, if for some sequence of normalizing constants $a_n > 0$ and b_n , $\Phi_{r:n}(a_n x + b_n) = P\left(\frac{1}{a_n}(X_{n-r+1:n} - b_n) \leq x\right)$ converges weakly to a nondegenerate limiting d.f. $\Phi_r(x)$, then $\Phi_r(x)$ must equal $1 - \Gamma_r(U(x))$, where $\Gamma_r(x) = \frac{1}{\Gamma(r)} \int_0^x t^{r-1} e^{-t} dt$ is the incomplete Gamma function, $\Gamma(r) = (r-1)!$ is the Gamma function and $U(x)$ has (up to location and scale changes) one and only one of the following three possible parametric forms, namely

Type I: $U_1(x) = e^{-x}$, $\forall x$;

Types II: $U_2^{(\alpha)}(x) = \begin{cases} \infty, & x \leq 0, \\ x^{-\alpha}, & \text{for some } \alpha > 0, x > 0; \end{cases}$

Types III: $U_3^{(\alpha)}(x) = \begin{cases} (-x)^\alpha, & \text{for some } \alpha > 0, x \leq 0; \\ 0, & x > 0. \end{cases}$

When $r = 1$, i.e., for maxima, the three “extreme value distributions” involved were discovered by Fréchet, and Fisher and Tippett, and discussed more completely later by Gnedenko (1943). When the rank r is any positive fixed integer they were completely discussed by Smirnov (1952). Let $\Phi_r(x)$ be a monotone function defined on a set $S \subseteq \mathfrak{R}_1$. By $\Phi_{r:n}(a_n x + b_n) \xrightarrow{S} \Phi_r(x)$, we denote restricted convergence of $\{\Phi_{r:n}\}$ on S , for all continuity points of S , that is, $\lim_{n \rightarrow \infty} \Phi_{r:n}(a_n x + b_n) = \Phi_r(x)$, for every continuity point $x \in S$ of Φ_r (isolated points of S are continuity points of Φ_r by convention). If $S = \mathfrak{R}_1$, $\Phi_{r:n} \xrightarrow{\mathfrak{R}_1} \Phi_r$, as $n \rightarrow \infty$ and $\mathcal{L}(\Phi_r) = \Phi_r(\infty) - \Phi_r(-\infty) = 1$, we write $\Phi_{r:n}(a_n x + b_n) \xrightarrow{w} \Phi_r(x)$, (complete convergence or weak convergence). The main problem of our investigation is the following:

If, for suitable normalizing constants $a_n > 0$ and b_n , we have the restricted convergence

$$\Phi_{r;n}(a_n x + b_n) \xrightarrow{S} \Phi_r(x), \text{ as } n \rightarrow \infty, \quad (1.1)$$

where $S \subset \mathfrak{R}_1$, under what conditions will the convergence in (1.1) continue, for all x , to an extreme value distribution (which clearly coincides with $\Phi_r(x)$ on S)? This problem is analogous to a problem that arose in the framework of a new theory of sums of independent r.v.'s which was stimulated by an idea of V.M. Zolotarev (see for example Riedel, 1977, and Rossberg and Siegel, 1981). The first result in this field was proved in Rossberg (1974). However, some results for this problem concerning the asymptotic theory of o.s. have been obtained by Gnedenko and Bereksey (1982a, 1983). More specifically the following two assertions are proved:

1. If the relation (1.1) holds, for some suitable normalizing constants $a_n > 0$ and b_n ; $r = 1$; $S = (c, d)$, where $c < 0$, $d > 0$ and $\Phi_1(d) - \Phi_1(c) > 0$ and $\Phi_1(x)$ is equal to one of the extreme value distributions, for all $x \in S$, then $\Phi_{1;n}(a_n x + b_n)$ converges weakly, for all x , to this extreme value distribution.
2. If the relation (1.1) holds for some suitable normalizing constants $a_n > 0$ and b_n ; $r = 1$; $S = \{x : x \leq A, A \text{ is a constant}\}$, where $\Phi_1(x)$ is any monotone function which has more than two growth points and $\Phi_1(-\infty) = 0$, then the convergence in (1.1) will continue, for all x to an extreme value distribution which coincides with $\Phi_1(x)$ on S .

When r is a positive fixed integer the assertion (2) is discussed in Gnedenko and Sherif (1983), while in Gnedenko, Barakat and Hemeda (1985) the assertion (1) is investigated for the joint d.f. of the r th and s th ($r < s$) with

the same original d.f. $F(x)$. Some important applications and interpretations “due to A.D. Solove’ev” of the above-stated results are given in Gnedenko and Bereksey (1982b). The survey paper by Rossberg (1995) stresses different points of the restricted convergence of d.f.’s of sums of independent r.v.’s and d.f.’s of o.s. In Barakat (1997), the assertion (1) is proved, for the r th extreme o.s., without the restrictive conditions “ $c < 0$, $d > 0$ ” and “ $\Phi_r(x)$ is equal to one of the extreme value distributions, for all $x \in (c, d)$.” Moreover, the same assertion is proved in Barakat (1997) (under some additional assumptions) for any central o.s. with rank r_n , for which $\sqrt{n} \left(\frac{r_n}{n} - \lambda \right) \rightarrow 0$, $\lambda \in (0, 1)$. In this paper the continuation property of the extreme o.s. ($r = 1, 2, \dots$) is investigated, when S is considered as a countable set of real values (set of measure zero). Such a continuation property is of obvious theoretical significance (see, e.g., Section 4) but it also has important practical applications, because the actual application of the extreme value distributions is that they provide in some cases exact but in most cases approximate probabilistic models for random quantities when the extremes govern the laws of interest (strength of material, floods, drought etc.). Therefore, a complicated situation can be handled by a comparatively simple asymptotic model. This can often be done by collecting a statistical sample on a random quantity and performing a goodness of fit test. However, there are often two major difficulties, which may restrict the advantage of this approach. The first difficulty is that the range of values for which the statistical samples are collected, is always limited. Therefore, the data actually enable us to identify the limit d.f. of the extreme o.s., which is of interest, only on a finite interval. The second difficulty is that many sampling procedures give only discrete statistical observations while many physical phenomena are (seemingly) continuous by

nature. However, these two difficulties may be considered as extrapolation and interpolation problems, respectively, and they become serious in some situations, for example in medical research and for agencies which regulate food or drug safety standards. As a further illustration we consider the following example.

Example (Extreme temperatures, see also Example 15.1.1 in Leadbetter *et al.*, 1983).

Experiments with thermometers and temperature measurements were started in a certain town during the month of July, around 1712–1713, shortly after Fahrenheit had constructed his temperature scale in 1709. These measurements became more regular and there is an almost complete series of daily temperature data from January 1739. During the first century of this period measurements were made only three or four times a day, and sometimes at irregular intervals. In 1839 observations of the true maxima and minima were made possible by installation of a max- and min-thermometer. Therefore, the data consists of $M_{1739}, \dots, M_{1838}$ (and $m_{1739}, \dots, m_{1838}$), where, e.g. M_i is the maximum of the discrete set of July temperatures in year i , and $M_{1839}, \dots, M_{1999}$ (and m_{1839}, m_{1999}), with M_j equal to the absolute (continuous) July maximum in year j . Evidently, all these observations actually enable us to identify the limit d.f. of maxima (and minima) only on the interval $[\underline{M}, \overline{M}]$ and $[\underline{m}, \overline{m}]$, where $\underline{M} = \min(M_{1739}, \dots, M_{1999})$ (and $\overline{M} = \max(M_{1739}, \dots, M_{1999})$) (and $\underline{m} = \min(m_{1739}, \dots, m_{1999})$, $\overline{m} = \max(m_{1739}, \dots, m_{1999})$). However, this difficulty (the first one) has been removed by the results of Barakat (1997). On the other hand, the discrete observations $M_{1739}, \dots, M_{1838}$ (and $m_{1739}, \dots, m_{1838}$), of course lead to some inaccuracy in the final decision about the choice of the limit d.f. of

maxima (and minima). This difficulty (the second difficulty) will be removed by Theorem 2.1 in this paper (another application of Theorem 2.1 will be given in Section 4). The methods of the proof of this theorem are different from the methods given in Gnedenko and Bereksey (1982a, 1983) and Barakat (1997). Moreover, the case when the sample size n becomes a positive integer valued r.v. is considered in Theorems 2.2, 2.3.

Some further conventions and notations. It will be convenient here to use the following notation: For any continuous d.f. $\Phi(x)$ and a positive fixed integer r , we write $\Phi \in T_r(I)$, if and only if there exist real constants $a > 0$ and b such that $\Phi(x) = 1 - \Gamma_r(U_1(ax + b))$; $\Phi \in T_r(II)$, if and only if there exist real constants $a, \alpha > 0$ and b such that $\Phi(x) = 1 - \Gamma_r(U_2^{(\alpha)}(ax + b))$; and $\Phi \in T_r(III)$, if and only if there exist real constants $a, \alpha > 0$ and b such that $\Phi(x) = 1 - \Gamma_r(U_3^{(\alpha)}(ax + b))$. (Note that the sets $T_r(I)$, $T_r(II)$ and $T_r(III)$ are disjoint.) Finally, we shall use the notation x_F , for the right endpoint of a d.f. F , i.e., $x_F = \sup \{x : F(x) < 1\}$, ($x_F \leq \infty$).

2. Main result

Theorem 2.1

Part 1: Let F be a continuous d.f. with $x_F = \infty$. Let $a_n > 0$ and b_n be suitable normalizing constants for which we have

$$\Phi_{r:n}(a_n x_i + b_n) \rightarrow l_i, \text{ for all } i = 1, 2, \dots, \text{ as } n \rightarrow \infty \quad (2.1)$$

where $\{x_i\}$ is an increasing sequence of real numbers such that $x_i \rightarrow \infty$, as $i \rightarrow \infty$ and the sequence $\{l_i\}$ has more than two different values in the interval $(0, 1)$. Then there exists a d.f. $\Phi_r \in T_r(I) \cup T_r(II)$, where $\Phi_r(x_i) = l_i$, $\forall i$, and

$$\Phi_{r:n}(a_n x + b_n) \xrightarrow{w} \Phi_r(x), \quad \forall x, \text{ as } n \rightarrow \infty. \quad (2.2)$$

Part 2: Let $x_F < \infty$ and $a_n > 0$, b_n be suitable normalizing constants for which the condition (2.1) is satisfied for a decreasing sequence $\{x_i\}$ of real values, where $x_i \rightarrow -\infty$, as $i \rightarrow \infty$. Then there exists a d.f. $\Phi_r \in T_r(I) \cup T_r(III)$, where $\Phi_r(x_i) = l_i$, $\forall i$, and the relation (2.2) is satisfied $\forall x$. When the sample size n is a positive integer valued r.v. ν_n and the interrelation of the basic variables $X_1, X_2, \dots$, and the random size ν_n is not restricted, we get the following extension of Theorem 2.1 in Barakat and Shandidy (1990) (for upper extremes) by using Theorem 2.1 in the present paper.

Theorem 2.2

Let $\frac{\nu_n}{n} \rightarrow \tau$, in probability, as $n \rightarrow \infty$, where τ is a positive r.v. Assume that there are sequences $a_n > 0$, b_n such that

$$\Phi_{r;n}(a_n x_i + b_n) \xrightarrow{S} l_i, \quad i = 1, 2, \dots, \quad \text{as } n \rightarrow \infty,$$

where the sequence $\{l_i\}$ has more than two different values in the interval $(0, 1)$ and $S = \{x_i\}$ is a sequence of real numbers such that $\{x_i\}$ is increasing, $x_i \rightarrow \infty$, as $i \rightarrow \infty$, if $x_F = \infty$ and $\{x_i\}$ is decreasing, $x_i \rightarrow -\infty$, as $i \rightarrow \infty$, if $x_F < \infty$. Then, as $n \rightarrow \infty$, there exists a monotone function $U(x)$ such that $\Gamma_r(U(x_i)) = 1 - l_i$, $i = 1, 2, \dots$, and

$$\Phi_{r;\nu_n}(a_n x + b_n) \xrightarrow{w} 1 - \int_0^\infty \Gamma_r(zU(x)) dP(\tau < z), \quad \text{as } n \rightarrow \infty.$$

Moreover,

$$1 - \Gamma_r(U(x)) \in T_r(I) \cup T_r(II), \quad \text{if } x_F = \infty,$$

and

$$1 - \Gamma_r(U(x)) \in T_r(I) \cup T_r(III), \quad \text{if } x_F < \infty.$$

Gnedenko and Gnedenko (1982) investigated the class of all possible limit d.f.'s of the normalizing sequences $\{X_{\nu_n:\nu_n}\} = \{\max(X_1, X_2, \dots, X_{\nu_n})\}$, when ν_n is assumed to be independent of all basic variables $X_1, X_2, \dots, X_n$, for any n and has a geometric d.f. with parameter $\frac{1}{n}$, i.e.,

$$P\left(\nu_n = \frac{1}{n}\right) = \frac{1}{n} \left(1 - \frac{1}{n}\right)^{k-1}, \quad k = 1, 2, \dots$$

This class contains only the following three parametric forms:

$$\Psi_1(x) = \frac{1}{1 + e^{-x}}, \quad \forall x \text{ (logistic d.f.)};$$

$$\Psi_2(x) = \begin{cases} 0, & x \leq 0, \\ \frac{x^\alpha}{1 + x^\alpha}, & x > 0, \text{ for some } \alpha > 0; \end{cases}$$

$$\Psi_3(x) = \begin{cases} 1, & x > 0, \\ \frac{1}{1 + (-x)^\alpha}, & x \leq 0, \text{ for some } \alpha > 0; \end{cases}$$

The following theorem gives an important extension of the above result and of the result of Gnedenko and Berekky (1982).

Theorem 2.3. Let ν_n be a r.v. which has a geometric d.f. with the parameter $\frac{1}{n}$. Let $a_n > 0$ and b_n be suitable normalizing constants for which

$$\Phi_{1:\nu_n}(a_n x_i + b_n) \rightarrow \frac{1}{1 + e^{-x_i}}, \quad i = 1, 2, \dots, n \rightarrow \infty,$$

where $\{x_i\}$ is an increasing sequence of real values such that $x_i \rightarrow \infty$, as $i \rightarrow \infty$, and $\{l_i\}$ has more than two different values in the interval $(0, 1)$.

Then

$$\Phi_{1:\nu_n}(a_n x + b_n) \xrightarrow{w} \Psi_1(x) = \frac{1}{1 + e^{-x}}, \quad \text{as } n \rightarrow \infty.$$

The above statement (with only the obvious changes) can be applied to Ψ_2 and Ψ_3 .

3. Proof

The proof of Theorem 2.1 will follow from a sequence of lemmas.

Lemma 3.1. For any two d.f.'s Φ_1 and Φ_2 we have $\Phi_1 \equiv \Phi_2$ if one of the following conditions is satisfied for some positive fixed integer r :

- (I) $\Phi_1, \Phi_2 \in T_r(I)$ and there exist two distinct points x_1 and x_2 such that $\Phi_1(x_i) = \Phi_2(x_i) = l_i$, $i = 1, 2$, where $0 < l_1 < l_2 < 1$.
- (II) $\Phi_1, \Phi_2 \in T_r(II)$ and there exists an increasing sequence of real numbers $\{x_i\}$ such that $x_i \rightarrow \infty$ as $i \rightarrow \infty$ and $\Phi_1(x_i) = \Phi_2(x_i)$, $\forall i = 1, 2, \dots$
- (III) $\Phi_1, \Phi_2 \in T_r(III)$ and there exists a decreasing sequence of real numbers $\{x_i\}$ such that $x_i \rightarrow -\infty$, as $i \rightarrow \infty$, and $\Phi_1(x_i) = \Phi_2(x_i)$, $\forall i = 1, 2, \dots$

Proof. Clearly, under the condition (I) there exist real constants $a > 0$ and b such that $\Phi_1(ax + b) = \Phi_2(x) = 1 - \Gamma_r(U_1(x))$, $\forall x$. In particular,

$$1 - \Gamma_r\left(e^{-(ax_i+b)}\right) = 1 - \Gamma_r\left(e^{-x_i}\right) = l_i, \quad i = 1, 2,$$

which yields $x_i = ax_i + b$, $i = 1, 2$ and $-\infty < x_1 < x_2 < \infty$ (since $0 < l_1 < l_2 < 1$). Hence, we deduce that $a = 1$ and $b = 0$, which means that $\Phi_1 \equiv \Phi_2$, as stated. If the condition (II) is satisfied, then there exist $\alpha_1, \alpha_2, a, c > 0$; b and d such that

$$\begin{aligned} \Phi_1(x) &= 1 - \Gamma_r\left(U_2^{(\alpha_1)}\left(\frac{x-b}{a}\right)\right), \quad \forall x, \text{ and} \\ \Phi_2(x) &= 1 - \Gamma_r\left(U_2^{(\alpha_2)}\left(\frac{x-d}{c}\right)\right), \quad \forall x. \end{aligned}$$

In particular, $\Phi_1(x_i) = \Phi_2(x_i)$, $\forall i = 1, 2, \dots$. Therefore,

$$U_2^{(\alpha_1)}\left(\frac{x_i - b}{a}\right) = U_2^{(\alpha_2)}\left(\frac{x_i - d}{c}\right), \quad \forall i = 1, 2, \dots$$

Hence, for sufficiently large i , we deduce that

$$\left[\frac{x_i - b}{a}\right]^{-\alpha_1} = \left[\frac{x_i - d}{c}\right]^{-\alpha_2}$$

which easily leads to

$$Y_i^\lambda = AY_i + B, \quad \text{for all sufficiently large } i, \quad (3.1)$$

where $Y_i = \frac{x_i - d}{c}$, $\lambda = \frac{\alpha_2}{\alpha_1} > 0$, $A = \frac{c}{a} > 0$ and $B = \frac{d - b}{a}$. For sufficiently large i , we can clearly choose a subsequence $\{x_{i_j}\}$ of $\{x_i\}$ such that $x_{i_1} < x_{i_2} < \dots < x_{i_j} < \dots$ and $\frac{x_{i_j}}{x_{i_1}} \rightarrow \infty$, as $j \rightarrow \infty$. Consequently, we have $Y_{i_1} < Y_{i_2} < \dots < Y_{i_j} < \dots$ and $\frac{Y_{i_j}}{Y_{i_1}} \rightarrow \infty$. On the other hand, the equations in (3.1) yield

$$A = \frac{[Y_{i_2}^\lambda - Y_{i_1}^\lambda]}{[Y_{i_2} - Y_{i_1}]} = \frac{[Y_{i_3}^\lambda - Y_{i_1}^\lambda]}{[Y_{i_3} - Y_{i_1}]} = \dots = \frac{[Y_{i_j}^\lambda - Y_{i_1}^\lambda]}{[Y_{i_j} - Y_{i_1}]} \quad (3.2)$$

Now we must have $\lambda = 1$. For, if not, we get, in view of (3.2),

$$A = \lim_{j \rightarrow \infty} \frac{Y_{i_j}^{\lambda-1} \left[1 - \left[\frac{Y_{i_1}}{Y_{i_j}} \right]^\lambda \right]}{\left[1 - \frac{Y_{i_1}}{Y_{i_j}} \right]} = \begin{cases} \infty, & \text{if } \lambda > 1, \\ 0, & \text{if } \lambda < 1, \end{cases}$$

(note that $Y_{i_j} \rightarrow \infty$, as $j \rightarrow \infty$) contradicting the fact that $0 < A < \infty$.

This, immediately leads to $A = 1$ (using (3.2)) and by virtue of (3.1) we deduce that $B = 0$, which means that $\alpha_1 = \alpha_2$, $a = c$ and $d = b$ or equivalently $\Phi_1 \equiv \Phi_2$ as required. The proof of the relation $\Phi_1 \equiv \Phi_2$, under

the condition (III) is similar to the proof under the condition (II) and for brevity it will be omitted. ■

Lemma 3.2. Let $0 \leq u \leq \infty$, and suppose that $a_n > 0$ and b_n are sequences of real numbers such that, for some point x_0 , $-\infty \leq x_0 \leq \infty$, we have

$$U_n(a_n x_0 + b_n) = n(1 - F(a_n x_0 + b_n)) \rightarrow u, \text{ as } n \rightarrow \infty. \quad (3.3)$$

Then

$$\Phi_{r:n}(a_n x_0 + b_n) \rightarrow l = 1 - \Gamma_r(u), \text{ as } n \rightarrow \infty, \quad (3.4)$$

where $0 \leq l \leq 1$. Conversely, if (3.4) holds for some fixed r , then (3.3) holds and so does (3.4) for all $r = 1, 2, \dots$

Proof. This lemma is a special case of Theorem 2.2.1 in Leadbetter *et al.* (1983). ■

Lemma 3.3. Let F be a continuous d.f. and $a_n > 0$, b_n suitable normalizing constants such that

$$\Phi_{r:n}(a_n x_i + b_n) \rightarrow l_i, \quad i = 1, 2, 3, \text{ as } n \rightarrow \infty, \quad (3.5)$$

where $-\infty < x_1 < x_2 < x_3 \leq \infty$ and $0 \leq l_1 < l_2 < l_3 \leq 1$. Then the sequence $\{\Phi_{r:n}(a_n x + b_n)\}_n$ is stochastically bounded (for the definition, see Feller, 1966).

Proof. The assertion of the lemma will follow, if we show that, for any subsequence $\{n_k\}$ of $\{n\}$, for which

$$\Phi_{r:n_k}(a_{n_k} x + b_{n_k}) \xrightarrow{\mathfrak{P}_1} \tilde{\Phi}_r(x), \text{ as } k \rightarrow \infty, \quad (3.6)$$

where $\tilde{\Phi}_r(x)$ is a nondecreasing right continuous function, we must have $\mathcal{L}(\tilde{\Phi}_r) = 1$, or equivalently $\tilde{\Phi}_r(\infty) = 1$ and $\tilde{\Phi}_r(-\infty) = 0$ (i.e., $\Phi_{r:n_k}(a_{n_k} x + b_{n_k}) \xrightarrow{w} \tilde{\Phi}_r(x)$). Clearly, $l_3 = 1$ immediately leads to

$\tilde{\Phi}_r(\infty) = 1$ while $l_1 = 0$ leads to $\tilde{\Phi}_r(-\infty) = 0$. Therefore we assume that $0 < l_1 < l_2 < l_3 < 1$. In view of Lemma 3.2, on the other hand, the limit relations (3.5) and (3.6) are equivalent to

$$U_n(a_n x_i + b_n) = n(1 - F(a_n x_i + b_n)) \rightarrow u_i \quad i = 1, 2, 3, \quad \text{as } n \rightarrow \infty,$$

and

$$U_{n_k}(a_{n_k} x + b_{n_k}) = n_k(1 - F(a_{n_k} x + b_{n_k})) \xrightarrow{\mathfrak{R}_1} \tilde{U}(x), \quad \text{as } k \rightarrow \infty, \quad (3.7)$$

respectively, where $1 - \Gamma_r(u_i) = l_i$, $0 < u_1 < u_2 < u_3 < \infty$, $1 - \Gamma_r(\tilde{U}(x)) = \tilde{\Phi}_r(x)$ and $\tilde{U}(x_i) = u_i$, $i = 1, 2, 3$. Hence, to prove the lemma, we have to prove $\tilde{U}(\infty) = 0$ and $\tilde{U}(-\infty) = \infty$. Now, for any positive real number t , the relation (3.6) yields (letting $[\cdot]$ denote integer part)

$$\Phi_{r:[n_k t]}(a_{[n_k t]} x + b_{[n_k t]}) \xrightarrow{\mathfrak{R}_1} \tilde{\Phi}_r(x), \quad \text{as } k \rightarrow \infty.$$

In view of Lemma 3.2, the last relation is equivalent to

$$U_{[n_k t]}(a_{[n_k t]} x + b_{[n_k t]}) \xrightarrow{\mathfrak{R}_1} \tilde{U}(x), \quad \text{as } k \rightarrow \infty.$$

In particular we have

$$U_{[n_k t]}(a_{[n_k t]} x_i + b_{[n_k t]}) \rightarrow u_i, \quad i = 1, 2, 3, \quad \text{as } k \rightarrow \infty.$$

Since, $0 < u_i < \infty$, $i = 1, 2, 3$, we have

$$1 - F(a_{[n_k t]} x_i + b_{[n_k t]}) \rightarrow 0, \quad i = 1, 2, 3, \quad \forall t > 0, \quad \text{as } k \rightarrow \infty.$$

Therefore,

$$U_{n_k}(a_{[n_k t]} x + b_{[n_k t]}) \xrightarrow{[x_1, x_3]} \frac{\tilde{U}(x)}{t} = U^*(x), \quad \forall t > 0, \quad \text{as } k \rightarrow \infty,$$

which again, in view of Lemma 3.2, yields

$$\Phi_{r:n_k}(a_{[n_k t]} x + b_{[n_k t]}) \xrightarrow{[x_1, x_3]} 1 - \Gamma_r(U^*(x)) = \Phi_r^*(x), \quad \forall t > 0. \quad (3.8)$$

Now, define the following sequences of truncated d.f.'s

$$F_{r:n}(x) = \begin{cases} 0, & x < \Phi_{r:n}^{-1}(l_1), \\ \frac{\Phi_{r:n}(x) - l_1}{l_3 - l_1}, & \Phi_{r:n}^{-1}(l_1) \leq x < \Phi_{r:n}^{-1}(l_3), \\ 1, & x \leq \Phi_{r:n}^{-1}(l_3); \end{cases}$$

and

$$\tilde{F}_r(x) = \begin{cases} 0, & x < \tilde{\Phi}_r^{-1}(l_1), \\ \frac{\tilde{\Phi}_r(x) - l_1}{l_3 - l_1}, & \tilde{\Phi}_r^{-1}(l_1) \leq x < \tilde{\Phi}_r^{-1}(l_3), \\ 1, & x \leq \tilde{\Phi}_r^{-1}(l_3); \end{cases}$$

where $F^{-1}(u) = \sup\{x : F(x) \leq u\}$ is the inverse of the cumulative d.f. F . Now, in view of (3.6), (3.8) and the conditions of our theorem it is not hard to verify that

$$F_{r:n_k}(a_{n_k}x + b_{n_k}) \xrightarrow{w} \tilde{F}_r(x), \text{ as } k \rightarrow \infty,$$

and

$$F_{r:n_k}(a_{[n_k t]}x + b_{[n_k t]}) \xrightarrow{w} F_r^*(x), \text{ as } k \rightarrow \infty,$$

where

$$F_r^*(x) = \begin{cases} 0, & x < \Phi_r^{*-1}(l_1), \\ \frac{\tilde{\Phi}_r^*(x) - l_1}{l_3 - l_1}, & \Phi_r^{*-1}(l_1) \leq x < \Phi_r^{*-1}(l_3), \\ 1, & x \leq \Phi_r^{*-1}(l_3). \end{cases}$$

Clearly, both $\tilde{F}_r$ and F_r^* are nondegenerate d.f.'s. Therefore, by virtue of Khintchine's convergence of types theorem, we deduce that there exist two functions $\alpha(t) > 0$ and $\beta(t)$ such that

$$F_r^*(\alpha(t)x + \beta(t)) = \tilde{F}_r(x), \quad \forall x.$$

Since, $0 < \tilde{F}_r(x_2) = \frac{l_2 - l_1}{l_3 - l_1} < 1$, then

$$\alpha(t)x_2 + \beta(t) \in \{x : \Phi_r^{*-1}(l_1) \leq x < \Phi_r^{*-1}(l_3)\}.$$

Moreover, we can easily deduce that

$$tu_2 = \tilde{U}(\alpha(t)x_2 + \beta(t)), \quad \forall t > 0. \quad (3.9)$$

Now, if the assumption $\tilde{U}(\infty) = 0$ is violated, we have, on account of (3.9), $u_2 = \frac{\tilde{U}(\alpha(t)x_2 + \beta(t))}{t} \geq \frac{\tilde{U}(\infty)}{t}$, $\forall t > 0$, contradicting the fact that $u_2 < \infty$ (let t be sufficiently small). And if $\tilde{U}(-\infty) < \infty$, we get, in view of (3.9), $\frac{\tilde{U}(-\infty)}{t} \geq \frac{\tilde{U}(\alpha(t)x_2 + \beta(t))}{t} = u_2$, $\forall t > 0$, contradicting the fact that $u_2 > 0$ (let t be sufficiently large) which proves that $\tilde{U}(\infty) = 0$ and $\tilde{U}(-\infty) = \infty$, as required to complete the proof. ■

Remark 3.1. It is possible to obtain a very short proof of Lemma 3.3, using only Theorem 3.1 in Barakat (1997), as follows:

Starting with (3.6) and the assumption $0 < l_1 < l_2 < l_3 < 1$, we get

$$\Phi_{r:n_k}(a_{n_k}x + b_{n_k}) \xrightarrow{[x_1^*, x_3^*]} \tilde{\Phi}_r(x), \quad \text{as } k \rightarrow \infty,$$

where x_1^* and x_3^* are any continuity points of $\tilde{\Phi}_r(x)$, such that $x_1 > x_1^*$ and $x_3 < x_3^*$. Clearly $\tilde{\Phi}_r(x)$ has at least two growth points in (x_1^*, x_3^*) (since $0 < \tilde{\Phi}_r(x_1) = l_1 < l_2 = \tilde{\Phi}_r(x_2) < l_3 = \tilde{\Phi}_r(x_3) < 1$). Therefore, on applying Theorem 3.1 in Barakat (1997) to the sequence $\{\Phi_{r:n_k}(a_{n_k}x + b_{n_k})\}$ we deduce that

$$\Phi_{r:n_k}(a_{n_k}x + b_{n_k}) \xrightarrow{w} \tilde{\Phi}_r(x), \quad \text{as } k \rightarrow \infty.$$

Moreover, $\tilde{\Phi}_r(x)$ is an extreme value distribution, i.e. $\mathcal{L}(\tilde{\Phi}_r) = 1$ (this of course does not ensure the convergence of the sequence $\{\Phi_{r:n}(a_nx + b_n)\}$). We now turn to the proof of Theorem 2.1. Clearly, in view of Lemma 3.3, if the relation (2.1) is satisfied, under the conditions of Part 1 or Part 2 of Theorem 2.1,

the sequence $\{\Phi_{r:n}(a_n x + b)\}_n$ will be stochastically bounded. Therefore, by virtue of Theorem 1, Chapter VII in Feller (1966), our theorem immediately follows if for any two subsequences $\{n_{k'}\}$ and $\{n_{k''}\}$ of $\{n\}$, for which

$$\Phi_{r:n_{k'}}(a_{n_{k'}}x + b_{n_{k'}}) \xrightarrow{w} \Phi'_r(x), \text{ as } k' \rightarrow \infty,$$

and

$$\Phi_{r:n_{k''}}(a_{n_{k''}}x + b_{n_{k''}}) \xrightarrow{w} \Phi''_r(x), \text{ as } k'' \rightarrow \infty,$$

we prove that $\Phi'_r(x) = \Phi''_r(x)$, $\forall x$. Clearly, if we have two such subsequences then, on account of the conditions of Parts 1 and 2, both limit d.f.'s Φ'_r and Φ''_r must be nondegenerate. Therefore, by virtue of the Extremal Types Theorem we have the following relations:

$$\Phi'_r, \Phi''_r \in T_r(I) \cup T_r(II), \text{ if } x_F = \infty,$$

$$\text{and } \Phi'_r, \Phi''_r \in T_r(I) \cup T_r(III), \text{ if } x_F < \infty.$$

Now, in view of Lemma 3.1, the proof of Part 1 of Theorem 2.1 will be complete if we prove that the two relations $\Phi'_r \in T_r(I)$ and $\Phi''_r \in T_r(II)$ cannot both hold. For, if both relations hold there exist real constants $a, c, \alpha > 0, b$ and d such that, for sufficiently large i , we have

$$\left[\frac{x_i - b}{a} \right]^{-\alpha} = e^{-\left(\frac{x_i - d}{c} \right)},$$

which is equivalent to

$$\left[1 + \frac{d - b}{cY_i} \right]^\alpha Y_i^\alpha e^{-Y_i} = \left[\frac{a}{c} \right]^\alpha,$$

where $Y_i = \frac{(x_i - d)}{c}$. Let $i \rightarrow \infty$ we get $\left(\frac{a}{c} \right)^\alpha = 0$ ($Y_i \rightarrow \infty$, as $i \rightarrow \infty$), which is impossible, as desired, to complete the proof of Part 1. The

remainder of the proof of Part 2 can be completed in a similar way and will be omitted for brevity. ■

Proof of Theorem 2.2. Using Theorem 2.1, there exists a nondegenerate d.f. Φ_r such that $\Phi_r(x_i) = l_i$, $\forall i$ and $\Phi_{r:n}(a_n x + b_n) \xrightarrow{w} \Phi_r(x)$, $\forall x$ as $n \rightarrow \infty$. Moreover, $\Phi_r \in T_r(I) \cup T_r(II)$, when $x_F = \infty$ and $\Phi_r \in T_r(I) \cup T_r(III)$, when $x_F < \infty$. Applying the Extremal Types Theorem, there exists a monotone function $U(x)$, for which $\Phi_r(x) = 1 - \Gamma_r(U(x))$. Consequently, Theorem 2.1 in Barakat and Shandidy (1990), is applicable and the proof is complete. ■

Proof of Theorem 2.3. First, we shall need the following lemma which can be proved in analogous fashion to the transfer theorem for maxima in Gnedenko and Gnedenko (1982).

Lemma. Let $a_n > 0$ and b_n be suitable normalizing constants such that

$$(1) \Phi_{1:n}(a_n x_0 + b_n) \rightarrow l, \text{ as } n \rightarrow \infty,$$

and

$$(2) P\left(\frac{\nu_n}{n} < x\right) \xrightarrow{w} A(x), \quad A(+0) = 0, \text{ where } A(x) \text{ is a d.f.}$$

Then

$$(3) \Phi_{1:\nu_n}(a_n x_0 + b_n) \rightarrow \int_0^\infty l^z dA(z), \text{ as } n \rightarrow \infty.$$

Clearly, under the conditions of Theorem 2.3, we have $A(x) = 1 - e^{-x}$, $x > 0$. Moreover, we can choose a subsequence $\{n_k\}$ of $\{n\}$ such that, for all $i = 1, 2, \dots$, we have

$$\Phi_{1:n_k}(a_{n_k} x_i + b_{n_k}) \rightarrow \bar{l}_i, \text{ as } k \rightarrow \infty.$$

In view of the preceding lemma, we get

$$\frac{1}{1 + e^{-x_i}} = \int_0^\infty \bar{l}_i^z d(1 - e^{-z}) = \int_0^\infty \left(\frac{\bar{l}_i}{e}\right)^z dz = \frac{1}{1 - \ln \bar{l}_i}, \quad i = 1, 2, \dots$$

which leads to $\bar{l}_i = \exp(-e^{-x_i})$, $i = 1, 2, \dots$ where, obviously, $\{\bar{l}_i\}$ has more than two different values in the interval $(0, 1)$. Therefore, our theorem immediately follows by virtue of Theorem 2.1 and the transfer theorem for maxima in Gnedenko and Gnedenko (1982). ■

4. Interpretation

The results of Section 2 can be theoretically interpreted in the following way. It is known that the three limiting extreme value distributions can be thought of as members of a single family of distributions, which is known as the generalized extreme value d.f. and has the form $\Phi_{r,c}(x) = 1 - \Gamma_r \left((1 + cx)^{-\frac{1}{c}} \right)$, $1 + cx > 0$, where c is an arbitrary real number. Thus, $\Phi_{r,c}(x)$, $c \neq 0$, is of the same type as $1 - \Gamma_r \left(U_2^{(\alpha)}(x) \right)$ or $1 - \Gamma_r \left(U_3^{(\alpha)}(x) \right)$, $\alpha = \left| \frac{1}{c} \right|$, depending on whether $c > 0$ or $c < 0$. For $c = 0$, $\Phi_{r,0}$ is defined as $\lim_{c \rightarrow 0} \Phi_{r,c} = 1 - \Gamma_r(U_1(x))$. The reason for having a single family $\Phi_{r,c}(x)$, for each $r = 1, 2, \dots$, of limiting distributions of extreme o.s. is the fact that a limiting distribution $\Phi_1(x)$ must satisfy the functional equation

$$\Phi_1^m(A_m x + B_m) = \Phi_1(x), \quad (4.1)$$

for all $m \geq 1$, where $A_m > 0$ and B_m are suitable constants (see, e.g. Arnold *et al.*, 1992). However, Theorem 2.1 generalizes the characterization of $\Phi_{1,c}(x) = \exp(-(1 + cx)^{-\frac{1}{c}})$, $1 + cx > 0$, by (4.1), by reducing the domain of (4.1) to a set of measure zero on which Φ_1 is strictly increasing. If (4.1) is valid on such a set, then Theorem 2.1 entails that (4.1) is valid on the whole real

line. This fact is of obvious theoretical significance because (4.1) is related to several branches of applied probability. For example, (4.1) with $m = 2$ and characteristic functions replacing the distribution function Φ_1 leads to the classical Cramer Theorem stating that normal distributions when decomposed into independent nondegenerate components are themselves normal. On the other hand, Theorem 2.1 can be practically interpreted in the following way. Let us consider a technical system composed from a large number of equivalent elements which fail independently. Suppose further that the connections are in parallel, which means that the system failure will take place with the failure of the element with the longest life time. Finally, suppose that the distributions of the life periods of the elements of the technical system are such that there exists a distribution approximating the real distribution of the system life. The results of our section (Theorem 2.1) could be interpreted in the following way. A large number of equivalent technical systems are tested at the discrete instants $T_1, T_2, \dots, T_i$, where T_i is large. As a result of the testing the distribution of the system life period at the moments $T_1, T_2, \dots, T_i$ is evaluated. In accordance with Theorem 2.1, the identification of the limit distribution type at the instants $T_1, T_2, \dots, T_i$ permits us to extend this relationship to the whole line.

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